

BARNWELL AGDM, AB, Canada

WMO: 713460

Lat: **49.8007N**Lon: **112.3026W**Elev: **2704**StdP: **13.32**Time Zone: **-7.00 (NAM)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.3	-13.5	-26.1	1.4	-19.3	-20.1	2.0	-13.3	35.7	43.1	31.6	39.4	5.9	230	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	89.3	63.1	85.7	62.0	82.3	61.3	66.8	82.1	64.8	80.2	63.0	78.2	9.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.4	89.9	73.8	59.1	82.8	71.0	57.0	76.6	68.9	32.9	82.1	31.3	80.6	29.8	78.3	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	27.9	24.0	DB	-28.5	95.2	5.6	3.8	-32.5	97.9	-35.8	100.1	-38.9	102.3	-43.0	105.0	(4)
(5)			WB	-28.8	71.9	5.3	2.6	-32.6	73.8	-35.7	75.3	-38.7	76.8	-42.6	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	21.9	21.3	30.9	42.5	52.2	59.7	65.9	63.8	55.3	43.6	30.9	21.1
	DBStd	19.79	16.02	15.62	14.93	8.97	8.18	5.63	5.09	5.85	8.09	9.84	13.73	15.35
	HDD50	4343	871	804	596	247	72	2	0	1	38	240	577	895
	HDD65	8358	1336	1224	1058	674	400	176	50	92	301	664	1023	1360
	CDD50	1627	0	0	3	24	141	293	493	430	198	41	4	0
	CDD65	165	0	0	0	0	4	17	78	55	11	0	0	0
	CDH74	3639	0	0	2	22	206	380	1421	1184	404	20	0	0
	CDH80	1233	0	0	0	3	48	88	514	443	136	1	0	0
Wind	WSAvg	10.5	11.9	10.2	11.2	11.6	10.9	10.1	8.6	8.3	9.5	10.8	11.6	11.3
Precipitation	PrecAvg	14.5	0.5	0.4	0.7	1.1	2.1	3.5	1.5	1.4	1.3	0.8	0.6	0.5
	PrecMax	20.2	1.1	0.9	2.4	2.4	4.1	8.9	4.6	3.1	4.3	2.6	1.4	1.1
	PrecMin	10.0	0.0	0.0	0.0	0.3	0.6	0.6	0.1	0.0	0.1	0.2	0.1	0.0
	PrecStd	2.7	0.3	0.2	0.5	0.5	0.9	2.0	1.1	0.8	1.0	0.6	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	56.7	65.5	77.4	85.2	87.7	93.6	92.7	89.9	77.0	64.4	53.6
		MCWB	44.5	42.4	47.5	52.6	58.6	64.7	64.7	62.8	59.9	54.1	49.2	41.7
	2%	DB	49.4	49.7	60.0	70.1	79.9	82.1	89.5	89.0	84.3	70.3	56.8	48.5
		MCWB	40.6	39.4	45.2	49.8	56.6	61.5	65.6	61.5	58.4	52.2	44.7	40.0
	5%	DB	45.5	45.4	55.4	64.7	74.8	78.5	85.7	85.1	78.5	64.6	52.6	44.7
		MCWB	37.9	36.8	42.3	47.0	54.2	60.1	64.3	60.9	56.4	49.6	42.6	37.0
	10%	DB	41.6	41.2	50.8	59.7	70.0	74.5	82.1	81.0	72.7	59.5	48.5	41.2
		MCWB	35.8	34.6	40.2	44.6	52.1	57.7	63.3	60.3	54.8	46.7	39.8	34.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	43.3	48.9	53.6	60.0	67.0	71.9	66.6	62.4	56.2	50.2	43.2
		MCDB	53.3	53.1	63.2	73.5	80.8	81.9	85.9	83.3	82.4	73.1	63.1	51.3
	2%	WB	40.9	39.9	46.0	50.6	57.2	63.9	68.4	64.4	59.6	53.1	45.9	40.2
		MCDB	49.0	49.1	58.2	68.4	76.7	78.3	83.5	81.6	79.6	68.0	55.6	47.7
	5%	WB	38.3	37.3	43.2	47.9	55.1	61.4	66.3	62.9	57.6	50.1	43.1	37.5
		MCDB	44.8	44.7	54.3	62.5	72.2	74.9	81.2	80.0	76.3	63.2	51.9	43.8
	10%	WB	36.0	34.7	40.4	45.5	53.0	59.1	64.5	61.2	55.4	47.6	40.3	35.0
		MCDB	41.4	40.9	49.9	58.2	67.4	71.8	79.3	77.8	71.1	58.4	47.8	40.9
Mean Daily Temperature Range	5% DB	MDBR	19.9	21.5	22.6	24.3	24.8	23.9	28.9	29.2	26.9	24.2	21.0	19.5
		MCDBR	20.7	25.0	29.0	35.0	34.1	31.4	35.7	38.4	38.0	32.1	25.1	20.5
		MCWBR	14.1	16.2	16.4	17.5	14.7	14.2	15.8	15.1	15.9	16.3	15.1	13.6
	5% WB	MCDBR	19.6	24.3	28.1	31.6	30.7	28.6	32.0	32.8	35.9	30.8	24.6	21.0
		MCWBR	13.6	16.6	16.7	16.5	13.8	14.1	15.8	14.3	15.8	16.1	15.7	14.6
Clear-Sky Solar Irradiance	taub	0.259	0.268	0.290	0.327	0.341	0.339	0.342	0.359	0.319	0.289	0.262	0.245	
	taud	2.454	2.482	2.473	2.375	2.383	2.437	2.443	2.370	2.456	2.519	2.497	2.445	
	Ebn at Noon	259	285	295	292	291	291	289	278	280	272	256	248	
	Edh at Noon	19	23	28	35	36	35	34	35	29	22	18	16	
All-Sky Solar Radiation	RadAvg	393	710	1116	1542	1835	2032	2227	1775	1261	784	437	308	
	RadStd	33	57	89	105	146	139	114	110	106	81	35	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page